

VINITH KUMAR T

AUTONOMOUS ROBOTICS ENGINEER

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Linkedin: linkedin.com/in/vinith-kumar-t-full-stack-robotics-engineer | Portfolio: vinithkumar-themorgan.github.io

PROFESSIONAL SUMMARY

Autonomous Robotics Engineer with 2+ years of professional experience taking robots from blank sheet to deployment, covering mechanical design and fabrication, electronics, embedded firmware, ROS/ROS 2 autonomy, system integration, and testing. Built 3 autonomous mobile robots (AMR) from scratch and developed navigation, SLAM, perception, and control stacks reaching approximately 10 mm to 20mm positional accuracy. Won a INR 1 crore government EdTech contract and earned recognition in six national robotics and drone competitions, including National Winner at KAVACH 2023 (Government of India). Seeking a multidisciplinary robotics role designing, integrating, and deploying complete autonomous systems.

TECHNICAL SKILLS

Robotics and Autonomy: ROS (Robot Operating System), ROS 2, Nav2 Navigation Stack, SLAM (Simultaneous Localization and Mapping), AMCL, Localization, Mapping, Odometry, Path Planning, Motion Planning, Obstacle Avoidance, Sensor Fusion, Forward and Inverse Kinematics, URDF, TF2, RViz, Gazebo, MuJoCo, Foxglove Studio, Autonomous Mobile Robots (AMR)

Perception, Computer Vision and AI: OpenCV, YOLO, Roboflow, Object Detection, Deep Learning, Machine Learning, Image Processing, LiDAR, Depth Cameras, Point Cloud Processing, ArUco Marker Detection

Control and State Estimation: PID Control, Multi-Controller Architecture, Trajectory Tracking, Closed-Loop Motor Control, Extended Kalman Filter (EKF), LiDAR-IMU Sensor Fusion, Differential Drive Kinematics

Programming and Development Tools: Python, C++, Linux, Ubuntu, Bash, Git, GitHub, Version Control, Debugging, Hardware-in-the-Loop Testing

Embedded Systems and Electronics: ESP32, STM32, Microcontroller Firmware, UART, I2C, SPI, CAN Bus, RS-485, EtherCAT, Encoder Integration, Motor Drivers, BMS (Battery Management System), Sensor and Actuator Integration, Circuit Design, Hardware Testing, Automated Test Jig Development

UAV, RF and Sensing: Pixhawk, PX4, ArduPilot, UAV and Drone Design, Autonomous Flight, Waveguide Design, RF Analysis, mmWave Sensing

Mechanical Design and Fabrication: SolidWorks, Onshape, CAD, 3D Printing, Rapid Prototyping, Mechanical Design, Gears and Transmissions, Sheet Metal Design, FEA (Finite Element Analysis), CFD (Computational Fluid Dynamics), MIG Welding, Laser Cutting, CNC Machining, Lathe and Milling, Design for Manufacturing (DFM)

Industrial Automation and IoT: MQTT, Node-RED, HMI Design, Control Panel Design and Wiring, Industrial Communication Protocols, System Integration

PROFESSIONAL EXPERIENCE

Robotics Engineer

Robolog Automation

Jun 2024 - Present

Tamil Nadu, India

- Built 3 autonomous mobile robots (AMR) end to end for EdTech and networking deployments, owning mechanical design, fabrication, electronics, embedded firmware, ROS 2 software, and system validation.
- Won a INR 1 crore Tamil Nadu government EdTech contract by demonstrating two EdTech AMR platforms (ROS and ROS 2) against multiple competing vendors in a competitive project allocation round.
- Architected a dynamic multi-controller navigation framework that selects control strategy by navigation objective, delivering approximately 10 mm to 20mm positional accuracy across path-following and goal-reaching tests.
- Developed and tuned complete ROS 2 autonomy stacks spanning SLAM mapping, localization, global and local path planning, and dynamic obstacle avoidance for indoor deployment.
- Designed and wired industrial control panels and built automated hardware test jigs that validate BMS circuits, motor drivers, and electronic modules, standardizing pre-deployment hardware validation.
- Drove the full product lifecycle from concept and CAD through fabrication, wiring, software integration, and final acceptance testing.

Robotics Engineer, Research and Development (Volunteer)

Mar 2024 - May 2024

Bannari Amman Institute of Technology

Tamil Nadu, India

- Programmed and integrated a ROS-based AMR for autonomous navigation, improving navigation speed by 20% and trajectory accuracy through LiDAR-IMU sensor fusion.
- Demonstrated the live autonomous navigation system to industry and academic audiences at INTEC Expo 2024.

Team Lead and Full-Stack Robotics Developer, Team Little Champs

Sep 2020 - May 2024

Bannari Amman Institute of Technology

Tamil Nadu, India

- Led a multidisciplinary robotics team from concept and development through system integration, testing, and national-level competitions.
- Developed autonomous mobile robots, UAVs, robotic manipulators, and AI/embedded systems, and mentored junior students in robotics and autonomous system development.

PROJECTS

Autonomous Security Mobile Robot

ROS, SLAM, Sensor Fusion, AI Perception, LiDAR

- Built an autonomous patrol robot integrating ROS navigation, SLAM, multi-sensor fusion, and AI-based perception for unmanned security and surveillance duty.

Autonomous Package Delivery Drone

Pixhawk, PX4, Computer Vision, ArUco, Depth Camera

- Developed an autonomous package-delivery UAV on Pixhawk using depth vision and ArUco-marker target detection for precision autonomous landing.

Six-Axis Vision-Guided Robotic Manipulator

Robotic Manipulation, Inverse Kinematics, Computer Vision, Automation

- Engineered a vision-guided six-axis manipulator for autonomous package identification, pick-and-place, and sorting. Patent applied.

Autonomous Mobile Manipulator

ROS, Raspberry Pi, Computer Vision, Mobile Manipulation

- Built a mobile manipulation platform pairing a robotic arm with vision-based autonomous navigation on an embedded Linux compute stack.

mmWave Vital-Sign Sensing and Waveguide System

RF, Waveguide Design, mmWave, Signal Processing

- Designed and analyzed a waveguide-based mmWave front end for non-contact vital-sign monitoring, evaluating propagation and RF performance.

Additional Projects: 7+ further projects spanning autonomous drones, mobile robotics, computer vision, embedded systems, industrial automation, RF/mmWave, and mechanical design. **Full portfolio:** [vinithkumar-themorgan.github.io](https://github.com/vinithkumar-themorgan)

HONORS AND AWARDS

- **KAVACH 2023** - National Winner, Government of India
- **Smart India Hackathon (SIH) 2022** - 2nd Runner-Up, Government of India
- **Flipkart GRID 5.0** - National Finalist, Robotics
- **Flipkart GRID 4.0** - Excellence Award, Drones
- **Patent Applied** - Six-Axis Pick-and-Place Robot
- **Publication** - Quadraped Robot, IRE Journals
- **Hackfest 2023** - Finalist, received internship offer
- **Aerathon 2023 (SAE)** - Ranked 23rd Nationally, Drones

EDUCATION

Bachelor of Engineering (B.E.), Electronics and Instrumentation Engineering

Sep 2020 - May 2024

Bannari Amman Institute of Technology, Tamil Nadu, India

CGPA: 7.9 / 10